Getting Started with CIMEHR

Introduction

Electronic Health Record (EHR) data are longitudinal but not collected by design. Patients visit at irregular times, and even when a visit occurs, an outcome or biomarker may not be measured. **CIMEHR** provides simple benchmark methods and semiparametric joint models for analyses affected by informative visiting and informative observation.

This vignette starts with the data format, then moves from simpler baseline methods to the more complex CIMEHR estimators. Examples are kept small (a 500-subject subset of the bundled dataset) so each chunk runs in a few seconds.

```
library(CIMEHR)
```

1. Input data format

The analysis functions expect **long-format data**, with one row per subject-visit or subject-time record. The CIMEHR estimators expect at least the following column roles, with default column names shown:

Column role	Default name	Meaning
Subject identifier	`id`	Unique subject identifier
Visit time	`time`	Visit time; `NA` indicates a row without a visit
Outcome	`Y`	Longitudinal outcome; usually `NA` when the outcome was not recorded
Observation indicator	`R`	1 if the outcome was recorded at the visit, 0 otherwise
Censoring/follow-up end	`C`	End of follow-up for each subject
Covariates	user-named	Baseline or model covariates

All package methods should be given long/panel data. The CIMEHR-family functions accept custom column names through `id_col`, `time_col`, `y_col`, `r_col`, and `censor_col`, so renaming your data is not required:

```
fit <- CIMEHR(
  data       = my_data,
  id_col     = "patient_id",
  time_col   = "visit_time",
  y_col      = "hba1c",
  r_col      = "measured",
  censor_col = "followup_end",
  covars_visit_XV              = c("age", "female"),
```

```
  covars_outcome_fixed_XY       = c("age", "female"),
  covars_outcome_random_link_ZY = "female",
  covars_obs_fixed_XO           = c("age", "female"),
  covars_obs_random_link_ZO     = "female"
)
```

1.1 Converting short/wide data to long data

If your dataset has one row per subject and repeated visit/outcome columns, reshape it first:

```
library(dplyr)
#>
#> Attaching package: 'dplyr'
#> The following objects are masked from 'package:stats':
#>
#>     filter, lag
#> The following objects are masked from 'package:base':
#>
#>     intersect, setdiff, setequal, union
library(tidyr)

wide_dat <- tibble::tibble(
  id = 1:2,
  Age = c(0.4, -0.2),
  Gender = c(1, 0),
  NSES = c(0.2, -0.4),
  C = c(60, 60),
  time_1 = c(5, 8),  time_2 = c(20, NA), time_3 = c(45, 50),
  Y_1 = c(1.2, NA), Y_2 = c(1.6, NA), Y_3 = c(2.0, 1.1)
)

long_dat <- wide_dat %>%
  pivot_longer(
    cols = matches("^(time|Y)_\\d+$"),
    names_to = c(".value", "visit"),
    names_pattern = "(time|Y)_(\\d+)"
  ) %>%
  mutate(R = as.integer(!is.na(Y))) %>%
  arrange(id, time)

head(long_dat)
#> # A tibble: 6 × 9
#>      id   Age Gender  NSES     C visit  time     Y     R
#>   <int> <dbl>  <dbl> <dbl> <dbl> <chr> <dbl> <dbl> <int>
#> 1     1   0.4      1   0.2    60 1         5   1.2     1
#> 2     1   0.4      1   0.2    60 2        20   1.6     1
#> 3     1   0.4      1   0.2    60 3        45   2       1
#> 4     2  -0.2      0  -0.4    60 1         8  NA       0
#> 5     2  -0.2      0  -0.4    60 3        50   1.1     1
#> 6     2  -0.2      0  -0.4    60 2        NA  NA       0
```

2. Example dataset

The package includes `sim_ehr_data`, a long-format simulated EHR cohort designed to look like an outpatient HbA1c extract. The bundled dataset has the following columns. Latent quantities used internally by the data-generating process (the shared factor, the visiting frailty, and per-process random effects) are not included, since they are not observable in real EHR data.

Column	Type	Description
`id`	integer	Subject identifier
`Age`	continuous	Standardized age (z-score from a uniform [18, 99] draw)
`Age_raw`	continuous	Age in years (the raw Uniform[18, 99] draw); for descriptive use only
`Gender`	binary	1 = male, 0 = female
`Marital`	binary	1 = married, 0 = not
`Black`	binary	1 = Black race, 0 = otherwise
`Hispanic`	binary	1 = Hispanic ethnicity, 0 = otherwise
`NSES`	continuous	Standardized neighborhood median household income
`time`	continuous	Visit time on [0, 60]; `NA` for subjects with no visits
`R`	binary	1 if HbA1c was recorded at the visit, 0 otherwise
`Y`	continuous	Standardized linear outcome from the data-generating model
`HbA1c`	continuous	Raw HbA1c (~ 4.0–9.0 range), `exp(log(5.6) + 0.05 * Y)`
`log_HbA1c`	continuous	Natural log of `HbA1c`; recommended outcome for analysis
`C`	continuous	End-of-follow-up time (60 for all subjects)

```
data("sim_ehr_data")
nrow(sim_ehr_data)
#> [1] 16626
head(sim_ehr_data)
#>   id       Age  Age_raw Gender Marital Black Hispanic     NSES      time R
#> 1  1 0.6988464 74.59255      1       0     0        1 1.028589  2.410759 1
#> 2  2 0.2040346 63.07897      0       0     0        0 0.875106  1.071746 1
#> 3  2 0.2040346 63.07897      0       0     0        0 0.875106  5.287477 1
#> 4  2 0.2040346 63.07897      0       0     0        0 0.875106  7.390875 1
#> 5  2 0.2040346 63.07897      0       0     0        0 0.875106 15.304416 1
#> 6  2 0.2040346 63.07897      0       0     0        0 0.875106 32.253425 1
#>            Y    HbA1c log_HbA1c  C
#> 1  0.4887859 5.738546  1.747206 60
#> 2 -3.2403006 4.762399  1.560752 60
#> 3 -3.7466355 4.643344  1.535435 60
#> 4 -5.3543535 4.284693  1.455049 60
#> 5 -5.3035319 4.295595  1.457590 60
#> 6 -3.5080981 4.699056  1.547362 60
```

For the rest of this vignette, we work with a 500-subject subset to keep the executable examples quick:

```
set.seed(1)
sub_ids <- sample(unique(sim_ehr_data$id), 500)
ehr500  <- sim_ehr_data[sim_ehr_data$id %in% sub_ids, ]
nrow(ehr500)
#> [1] 4086
length(unique(ehr500$id))
#> [1] 500
```

The true simulation parameters are stored as an attribute and can be passed to `method_comparisons()` to populate true-value, bias, and RMSE columns:

```
true_params <- attr(sim_ehr_data, "true_params")
true_params$beta
#> intercept       Age    Gender   Marital     Black  Hispanic      NSES
#>     -2.00      0.05      0.10     -0.03      0.15      0.05     -0.25
true_params$gamma
#> intercept       Age    Gender   Marital     Black  Hispanic      NSES
#>     -2.00      0.10     -0.10     -0.05      0.15     -0.05      0.12
```

3. Method comparison

This table shows a comprehensive comparison of statistical methods for longitudinal EHR data. Methods are differentiated by whether they account for informative visiting (IP) and informative observation (IO), the specific scope of the shared latent structure linking these processes to the outcome, and the adjustment mechanism applied. Symbols: ✓ = explicitly handled or modeled; ✗ = ignored or not handled.

Abbreviations: LMM, linear mixed model; VA/OA, visit-adjusted / observation-adjusted; JMVL, joint modeling of visiting and longitudinal data; IIRR, inverse intensity rate ratio; MAR, missing at random; MI, multiple imputation; LI, linear increment.

Outcome-only

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
Summary statistics	✗	✗	None	Reduces the longitudinal series to a single scalar (e.g., min, mean, median, max) for regression.	—
Standard LMM (Laird and Ware, 1982)	✗	✗	None	Conditions on observed covariates	lme4

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
				(valid only under MAR).	
VA-LMM (Goldstein et al., 2016)	✓	✗	None	Includes *visit count* as a fixed covariate proxy.	lme4
OA-LMM	✗	✓	None	Includes *observation count* as a fixed covariate proxy.	lme4

IP-only

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
Liang (Liang et al., 2009)	✓	✗	Visiting & Outcome	Joint likelihood estimation via shared frailty linking visiting and outcome.	IrregLong
JMVL-G (Gasparini et al., 2020)	✓	✗	Visiting & Outcome	Joint likelihood modeling inter-visit times via shared random effects.	merlin
IIRR-Weighting (Bůžková and Lumley, 2007)	✓	✗	None	Inverse intensity weighting based on visiting intensity.	—
IIRR-Balanced (Yiu and Su, 2025)	✓	✗	None	Inverse intensity weighting adjusted for covariate balance.	—
Pairwise Likelihood	✓	✗	None	Constructs a likelihood from all within-	—

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
(Chen et al., 2015)				subject pairs of observations; conditioning on observed visit times eliminates the visit intensity from the likelihood.	

Imputation + IP-only

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
Multiple Imputation (MI) (Rubin, 1987)	✓	✓	Determined by IP-only	Multiple imputation based on posterior draws, followed by IP-only methods.	mice
Linear Increment Methods (LI) (Aalen and Gunnes, 2010)	✓	✓	Determined by IP-only	Deterministic linear interpolation between observed visits to impute missing values, followed by IP-only methods.	slim

IP+IO

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
Adapted-Liang (Liang et al., 2009)	✓	✓	Visiting & Outcome	Joint model structure defined, but observation coefficients are	—

Method	IP	IO	Shared Latent	Adjustment Mechanism	Package
				constrained to zero.	
EHRJoint (Du et al., 2025)	✓	✓	Visiting & Outcome	Tripartite model; observation process modeled via covariates only (no latent link).	—
Proposed (this package)	✓	✓	All three processes	Tripartite model; full shared latent structure links Visiting, Observation, and Outcome.	CIMEHR

IP = informative visiting / informative presence adjustment; IO = informative observation adjustment. The "Proposed" row corresponds to `CIMEHR()`; `CIMEHR_timevarying_integral()` and `CIMEHR_timevarying_ou()` extend this framework with more flexible observation-process modeling.

4. Simpler benchmark methods first

We define a covariate vector once and reuse it. The focal coefficient for the bundled dataset is `NSES`.

```
covars_all <- c("Age", "Gender", "Marital", "Black", "Hispanic", "NSES")
out_form   <- log_HbA1c ~ Age + Gender + Marital + Black + Hispanic + NSES + time
```

4.1 Summary-statistic regression

`Summary_stat()` is a naive baseline that collapses each subject's observed outcomes to a summary statistic and fits a cross-sectional regression.

```
fit_ss <- Summary_stat(
  ehr500,
  outcome = "log_HbA1c",
  formula = ~ Age + Gender + Marital + Black + Hispanic + NSES
)
summary(fit_ss)
#>
#> Summary-Statistic Regression -- Summary
#> ========================================================
#>
#> Fixed Effect (Coefficients):
#> --------------------------------------------------------
```

```
#>             Estimate Std. Error  t value Pr(>|t|)
#> (Intercept)   1.6385     0.0078 209.6145   0.0000
#> Age           0.0043     0.0044   0.9933   0.3211
#> Gender        0.0240     0.0086   2.7767   0.0057
#> Marital      -0.0162     0.0089  -1.8118   0.0707
#> Black        -0.0136     0.0091  -1.5042   0.1333
#> Hispanic      0.0090     0.0106   0.8526   0.3944
#> NSES          0.0007     0.0045   0.1500   0.8808
```

4.2 Linear mixed model

`Linear_mixed_model()` wraps `nlme::lme()` and supports standard, observation-adjusted, and visit-adjusted variants.

```
fit_lmm <- Linear_mixed_model(
  ehr500,
  fixed         = out_form,
  random        = ~ 1 | id,
  obs_indicator = "R"
)
summary(fit_lmm)
#>
#> Linear Mixed Model -- Summary
#> ============================================================
#>
#> Fixed Effect
#> ------------------------------------------------------------
#>               Value Std.Error   DF  t-value p-value  CI.low CI.high
#> (Intercept)  1.6212    0.0080 1874 202.9421  0.0000  1.6056  1.6369
#> Age          0.0049    0.0043  414   1.1490  0.2512 -0.0035  0.0133
#> Gender       0.0232    0.0085  414   2.7268  0.0067  0.0065  0.0400
#> Marital     -0.0155    0.0088  414  -1.7639  0.0785 -0.0327  0.0017
#> Black       -0.0106    0.0089  414  -1.1944  0.2330 -0.0281  0.0068
#> Hispanic     0.0080    0.0104  414   0.7635  0.4456 -0.0125  0.0284
#> NSES         0.0022    0.0045  414   0.4847  0.6282 -0.0066  0.0109
#> time         0.0006    0.0001 1874   8.1695  0.0000  0.0004  0.0007
```

A visit-adjusted variant adds the running visit count as a fixed covariate before observation-indicator filtering:

```
fit_va <- Linear_mixed_model(
  ehr500,
  fixed         = out_form,
  random        = ~ 1 | id,
  obs_indicator = "R",
  visit_adjust  = "VA"
)
summary(fit_va)
#>
#> Linear Mixed Model -- Summary
#> ============================================================
```

```
#>
#> Fixed Effect
#> ------------------------------------------------------------
#>               Value Std.Error   DF  t-value p-value  CI.low CI.high
#> (Intercept)  1.6218    0.0081 1873 201.2763  0.0000  1.6060  1.6376
#> Age          0.0048    0.0043  414   1.1268  0.2605 -0.0036  0.0132
#> Gender       0.0233    0.0085  414   2.7349  0.0065  0.0066  0.0400
#> Marital     -0.0155    0.0088  414  -1.7607  0.0790 -0.0327  0.0018
#> Black       -0.0106    0.0089  414  -1.1937  0.2333 -0.0281  0.0068
#> Hispanic     0.0079    0.0104  414   0.7582  0.4487 -0.0125  0.0284
#> NSES         0.0022    0.0045  414   0.4895  0.6247 -0.0066  0.0109
#> time         0.0005    0.0001 1873   5.3924  0.0000  0.0003  0.0007
#> n_visits     0.0001    0.0002 1873   0.5792  0.5625 -0.0002  0.0004
```

4.3 Pairwise likelihood

`Pairwise_likelihood()` fits a pairwise composite likelihood model for observed outcomes. The outcome column is read from `y_col`, so we set `y_col = "log_HbA1c"` explicitly here (the default is `"Y"`).

```
obs_data <- ehr500[!is.na(ehr500$R) & ehr500$R == 1, ]

fit_pl <- Pairwise_likelihood(
  data        = obs_data,
  formula     = out_form,
  y_col       = "log_HbA1c",
  pair_sample = 1000
)
summary(fit_pl)
#>
#> Pairwise Likelihood -- Summary
#> ============================================================
#>
#>   Observations: 2296 | Clusters: 421 | Pairs: 1000
#>   Converged: TRUE | Log-likelihood: -307.09
#>
#> Fixed Effect (Coefficients)
#> ------------------------------------------------------------
#>           Estimate Std.Error  CI.low CI.high
#> Age         0.0054    0.0041 -0.0025  0.0134
#> Gender      0.0343    0.0086  0.0173  0.0512
#> Marital    -0.0203    0.0087 -0.0374 -0.0031
#> Black      -0.0039    0.0084 -0.0204  0.0125
#> Hispanic   -0.0194    0.0103 -0.0395  0.0007
#> NSES        0.0221    0.0040  0.0144  0.0299
#> time        0.0007    0.0002  0.0002  0.0012
```

5. Visiting-process and joint methods

5.1 Inverse intensity rate ratio weighting

```
fit_iirr <- Inverse_intensity_rate_ratio_weighting(
  ehr500,
  outcome    = "log_HbA1c",
  visit_covs = covars_all
)
summary(fit_iirr)
#>
#> IIRR Weighting Estimator -- Summary
#> ============================================================
#>
#> Note: Standard errors are not available for this method.
#> Consider bootstrap for inference.
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>          Estimate
#> Age        0.2023
#> Gender    -0.1832
#> Marital   -0.0881
#> Black     -0.0032
#> Hispanic   0.0195
#> NSES      -0.1823
#>
#> Longitudinal Outcome -- Fixed Effect (beta)
#> ------------------------------------------------------------
#>             Estimate
#> (Intercept)   1.6183
#> Age           0.0057
#> Gender        0.0393
#> Marital      -0.0266
#> Black        -0.0093
#> Hispanic     -0.0065
#> NSES          0.0170
#> time          0.0008
```

5.2 Inverse intensity rate ratio balancing

`Inverse_intensity_rate_ratio_balancing()` extends IIRR with covariate-balancing weights. The `balance_covs` argument defaults to `visit_covs` when omitted. The `phi` argument controls a sensitivity tuning parameter.

```
fit_iirr_bal <- Inverse_intensity_rate_ratio_balancing(
  ehr500,
  outcome    = "log_HbA1c",
  visit_covs = covars_all
)
#> Note: initial value of fn function contains non-finite values (starting at index=2)
```

```
#>   Check initial x and/or correctness of function. Check initial x and/or correctness of
       function. Falling back to Maximum Likelihood Estimation (MLE) weights.
summary(fit_iirr_bal)
#>
#> IIRR Balancing-Weights Estimator -- Summary
#> ============================================================
#>
#> Note: Standard errors are not available for this method.
#> Consider bootstrap for inference.
#>
#>   Sensitivity parameter (phi): 0.0000
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>          Estimate
#> Age             0
#> Gender          0
#> Marital         0
#> Black           0
#> Hispanic        0
#> NSES            0
#>
#> Outcome -- Fixed Effect (beta) -- Balancing Weights
#> ------------------------------------------------------------
#>             Estimate
#> (Intercept)   1.6210
#> Age           0.0011
#> Gender        0.0451
#> Marital      -0.0357
#> Black         0.0025
#> Hispanic     -0.0241
#> NSES          0.0216
#> time          0.0007
#>
#> Outcome -- Fixed Effect (beta) -- MLE Weights
#> ------------------------------------------------------------
#>             Estimate
#> (Intercept)   1.6210
#> Age           0.0011
#> Gender        0.0451
#> Marital      -0.0357
#> Black         0.0025
#> Hispanic     -0.0241
#> NSES          0.0216
#> time          0.0007
```

5.3 Liang–Lu–Ying joint visiting/outcome model

`Joint_modeling_visiting_and_longitudinal_Liang()` fits a joint model for the visiting process and the longitudinal outcome, returning model-based standard errors for the visiting stage and a clustered sandwich estimator for the outcome stage (conditional on the Stage 1 nuisance estimates).

```
fit_liang <- Joint_modeling_visiting_and_longitudinal_Liang(
  ehr500,
  outcome     = "log_HbA1c",
  visit_covs  = covars_all,
  long_covs   = covars_all,
  random_covs = "NSES"
)
summary(fit_liang)
#>
#> JMVL (Liang) Joint Model -- Summary
#> ============================================================
#>
#> Note: Standard errors are not available for this method.
#> Consider bootstrap for inference.
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>          Estimate
#> Age        0.2023
#> Gender    -0.1832
#> Marital   -0.0881
#> Black     -0.0032
#> Hispanic   0.0195
#> NSES      -0.1823
#>
#>   Frailty variance: 1.4792
#>
#> Longitudinal Outcome -- Fixed Effect (beta)
#> ------------------------------------------------------------
#>          Estimate
#> Age        0.0051
#> Gender     0.0379
#> Marital   -0.0248
#> Black     -0.0011
#> Hispanic  -0.0061
#> NSES      -0.0100
```

5.4 EHRJoint

`EHRJoint()` is a tripartite joint model. Its observation process uses a Generalized Linear Model (GLM) with a probit link by default (set `obs_link = "logit"` to switch). The fitted object returns model-based standard errors for the visiting and observation stages and a clustered sandwich estimator for the outcome stage.

```
fit_ej <- EHRJoint(
  ehr500,
  outcome     = "log_HbA1c",
  visit_covs  = covars_all,
  long_covs   = covars_all,
  random_covs = "NSES"
)
```

```
summary(fit_ej)
#>
#> EHRJoint Model -- Summary
#> ============================================================
#>
#> Note: Standard errors are not available for this method.
#> Consider bootstrap for inference.
#>
#> Model note: visiting process uses a visit-count/frailty-compensation component.
#> Observation process uses logistic regression with a logit link.
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>          Estimate
#> Age        0.1910
#> Gender    -0.0954
#> Marital   -0.1797
#> Black     -0.0037
#> Hispanic   0.1481
#> NSES       0.0281
#>
#>   Frailty variance: 1.4502
#>
#> Observation Process -- Fixed Effect (alpha)
#> ------------------------------------------------------------
#>             Estimate
#> (Intercept)   0.2750
#> Age           0.0978
#> Gender       -0.1319
#> Marital       0.2160
#> Black         0.1318
#> Hispanic     -0.2345
#> NSES         -0.3036
#>
#> Longitudinal Outcome -- Fixed Effect (beta)
#> ------------------------------------------------------------
#>          Estimate
#> Age       -0.0151
#> Gender     0.0820
#> Marital   -0.1397
#> Black      0.0146
#> Hispanic  -0.2048
#> NSES       0.2355
#>
#> Longitudinal Outcome -- Random Link (theta)
#> ------------------------------------------------------------
#>   Estimate
#> 1  -0.1850
#> 2  -0.2012
```

6. Primary CIMEHR methods

6.1 Base CIMEHR

`CIMEHR()` jointly models informative visiting, informative observation, and the longitudinal outcome. Its visiting process uses Andersen–Gill partial likelihood with log-normal frailty, and its observation process uses a probit model with latent-random-link random effects when specified.

```
fit_cimehr <- CIMEHR(
  data                          = ehr500,
  y_col                         = "log_HbA1c",
  covars_visit_XV               = covars_all,
  covars_outcome_fixed_XY       = covars_all,
  covars_outcome_random_link_ZY = "NSES",
  covars_obs_fixed_XO           = covars_all,
  covars_obs_random_link_ZO     = "NSES"
)
print(fit_cimehr) # Print function only prinnts the outcome process estimates
#>
#> CIMEHR -- Outcome Fixed Effect (beta)
#> ================================================
#>           Estimate Std.Error  CI.low CI.high
#> Age         0.0040    0.1167 -0.2247  0.2327
#> Gender      0.0499    0.2317 -0.4043  0.5041
#> Marital    -0.0391    0.2469 -0.5231  0.4449
#> Black       0.0085    0.2360 -0.4541  0.4710
#> Hispanic   -0.0115    0.2977 -0.5949  0.5720
#> NSES        0.0047    0.1308 -0.2516  0.2611
#>
#> Use summary() for all process estimates, or
#> summary(x, specify = "visiting" / "observation" / "outcome") for one layer.
summary(fit_cimehr)
#>
#> CIMEHR Joint Model -- Summary
#> ============================================================
#>
#> Model note: visiting process uses Andersen-Gill partial likelihood with log-normal frailty
        where estimated.
#> Observation process uses a probit link; the OU variant adds OU serial correlation via PCL.
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>           Estimate Std.Error  CI.low CI.high
#> Age         0.1940    0.0604  0.0756  0.3123
#> Gender     -0.0976    0.1111 -0.3153  0.1201
#> Marital    -0.1850    0.1155 -0.4114  0.0414
#> Black      -0.0012    0.1173 -0.2312  0.2288
#> Hispanic    0.1480    0.1410 -0.1283  0.4244
#> NSES        0.0314    0.0657 -0.0974  0.1601
#>
```

```
#>   Frailty variance (sigma^2_zeta): 0.8944
#>
#> Observation Process -- Fixed Effect (alpha)
#> ------------------------------------------------------------
#>             Estimate Std.Error  CI.low CI.high
#> (Intercept)   0.4296    0.0904  0.2524  0.6067
#> Age           0.0904    0.0430  0.0062  0.1746
#> Gender       -0.0587    0.0851 -0.2254  0.1080
#> Marital       0.1235    0.0893 -0.0516  0.2986
#> Black         0.1030    0.0844 -0.0624  0.2685
#> Hispanic     -0.1292    0.1025 -0.3302  0.0717
#> NSES          0.3054    0.0688  0.1706  0.4401
#>
#> Observation Process -- Random Link (delta)
#> ------------------------------------------------------------
#>             Estimate Std.Error  CI.low CI.high
#> (Intercept)  -0.2788    0.0425 -0.3620 -0.1956
#> NSES         -0.5454    0.0513 -0.6458 -0.4449
#>
#>   Random-effect SD (Sigma_q):
#> (Intercept)        NSES
#>      0.2191      0.1109
#>
#> Longitudinal Outcome -- Fixed Effect (beta)
#> ------------------------------------------------------------
#>          Estimate Std.Error  CI.low CI.high
#> Age        0.0040    0.1167 -0.2247  0.2327
#> Gender     0.0499    0.2317 -0.4043  0.5041
#> Marital   -0.0391    0.2469 -0.5231  0.4449
#> Black      0.0085    0.2360 -0.4541  0.4710
#> Hispanic  -0.0115    0.2977 -0.5949  0.5720
#> NSES       0.0047    0.1308 -0.2516  0.2611
#>
#> Longitudinal Outcome -- Random Link (theta)
#> ------------------------------------------------------------
#>           Estimate Std.Error  CI.low CI.high
#> Intercept   0.0219    0.1557 -0.2833  0.3271
#> NSES        0.0263    0.1532 -0.2740  0.3267
```

6.1.1 Stage-specific summaries and coefficient extraction

You can print one process at a time and extract a particular coefficient programmatically:

```
summary_visiting(fit_cimehr)
#>
#> CIMEHR Joint Model -- Summary
#> ============================================================
#>
#> Model note: visiting process uses Andersen-Gill partial likelihood with log-normal frailty
       where estimated.
```

```
#> Observation process uses a probit link; the OU variant adds OU serial correlation via PCL.
#> Layer requested: visiting
#>
#> Visiting Process -- Fixed Effect (gamma)
#> ------------------------------------------------------------
#>          Estimate Std.Error  CI.low CI.high
#> Age        0.1940    0.0604  0.0756  0.3123
#> Gender    -0.0976    0.1111 -0.3153  0.1201
#> Marital   -0.1850    0.1155 -0.4114  0.0414
#> Black     -0.0012    0.1173 -0.2312  0.2288
#> Hispanic   0.1480    0.1410 -0.1283  0.4244
#> NSES       0.0314    0.0657 -0.0974  0.1601
#>
#>   Frailty variance (sigma^2_zeta): 0.8944
summary_observation(fit_cimehr)
#>
#> CIMEHR Joint Model -- Summary
#> ============================================================
#>
#> Model note: visiting process uses Andersen-Gill partial likelihood with log-normal frailty
      where estimated.
#> Observation process uses a probit link; the OU variant adds OU serial correlation via PCL.
#> Layer requested: observation
#>
#> Observation Process -- Fixed Effect (alpha)
#> ------------------------------------------------------------
#>             Estimate Std.Error  CI.low CI.high
#> (Intercept)   0.4296    0.0904  0.2524  0.6067
#> Age           0.0904    0.0430  0.0062  0.1746
#> Gender       -0.0587    0.0851 -0.2254  0.1080
#> Marital       0.1235    0.0893 -0.0516  0.2986
#> Black         0.1030    0.0844 -0.0624  0.2685
#> Hispanic     -0.1292    0.1025 -0.3302  0.0717
#> NSES          0.3054    0.0688  0.1706  0.4401
#>
#> Observation Process -- Random Link (delta)
#> ------------------------------------------------------------
#>             Estimate Std.Error  CI.low CI.high
#> (Intercept)  -0.2788    0.0425 -0.3620 -0.1956
#> NSES         -0.5454    0.0513 -0.6458 -0.4449
#>
#>   Random-effect SD (Sigma_q):
#> (Intercept)        NSES
#>      0.2191      0.1109
summary_outcome(fit_cimehr)
#>
#> CIMEHR Joint Model -- Summary
#> ============================================================
#>
#> Model note: visiting process uses Andersen-Gill partial likelihood with log-normal frailty
      where estimated.
#> Observation process uses a probit link; the OU variant adds OU serial correlation via PCL.
```

```
#> Layer requested: outcome
#>
#> Longitudinal Outcome -- Fixed Effect (beta)
#> ------------------------------------------------------------
#>          Estimate Std.Error  CI.low CI.high
#> Age        0.0040    0.1167 -0.2247  0.2327
#> Gender     0.0499    0.2317 -0.4043  0.5041
#> Marital   -0.0391    0.2469 -0.5231  0.4449
#> Black      0.0085    0.2360 -0.4541  0.4710
#> Hispanic  -0.0115    0.2977 -0.5949  0.5720
#> NSES       0.0047    0.1308 -0.2516  0.2611
#>
#> Longitudinal Outcome -- Random Link (theta)
#> ------------------------------------------------------------
#>           Estimate Std.Error  CI.low CI.high
#> Intercept   0.0219    0.1557 -0.2833  0.3271
#> NSES        0.0263    0.1532 -0.2740  0.3267
```

```
extract_coefficient(fit_cimehr, stage = "outcome",     parameter = "NSES")
#>        NSES
#> 0.004748068
extract_coefficient(fit_cimehr, stage = "visiting",    parameter = "NSES")
#>       NSES
#> 0.03135098
extract_coefficient(fit_cimehr, stage = "observation", parameter = "NSES")
#>      NSES
#> 0.3053833
```

6.2 CIMEHR with Gauss–Hermite quadrature

`CIMEHR_timevarying_integral()` uses Gauss–Hermite quadrature and online filtering for a more flexible observation process. Analytic standard errors are not currently exposed for this variant, so bootstrap inference is recommended (see Section 9).

```
fit_integral <- CIMEHR_timevarying_integral(
  data                          = ehr500,
  y_col                         = "log_HbA1c",
  covars_visit_XV               = covars_all,
  covars_outcome_fixed_XY       = covars_all,
  covars_outcome_random_link_ZY = "NSES",
  covars_obs_fixed_XO           = covars_all,
  covars_obs_random_link_ZO     = "NSES",
  gh_points_U = 7,
  gh_points_q = 5
)
```

6.3 CIMEHR with OU pairwise composite likelihood

`CIMEHR_timevarying_ou()` models serial correlation in the observation process using an Ornstein–Uhlenbeck process and pairwise composite likelihood.

```
fit_ou <- CIMEHR_timevarying_ou(
  data                          = ehr500,
  y_col                         = "log_HbA1c",
  covars_visit_XV               = covars_all,
  covars_outcome_fixed_XY       = covars_all,
  covars_outcome_random_link_ZY = "NSES",
  covars_obs_fixed_XO           = covars_all,
  covars_obs_random_link_ZO     = "NSES",
  pair_type = "adjacent"
)
```

7. Comparing methods with `method_comparisons()`

The helper `method_comparisons()` fits any user-specified combination of methods and returns separate tables for the visiting, observation, and outcome processes. If a method does not estimate a given process, that entry is shown as `NA`.

```
available_comparison_methods()
#>  [1] "CIMEHR"
#>  [2] "CIMEHR_timevarying_integral"
#>  [3] "CIMEHR_timevarying_ou"
#>  [4] "EHRJoint"
#>  [5] "Inverse_intensity_rate_ratio_balancing"
#>  [6] "Inverse_intensity_rate_ratio_weighting"
#>  [7] "Joint_modeling_visiting_and_longitudinal_Liang"
#>  [8] "Linear_increment"
#>  [9] "Linear_mixed_model"
#> [10] "Multiple_imputation"
#> [11] "Pairwise_likelihood"
#> [12] "Summary_stat"
```

The full call below fits five methods at once on the 500-subject subset. It takes a minute or two to run; if you are building the vignette and want to skip the runtime, change `eval = TRUE` to `eval = FALSE` in the chunk header.

```
cmp_subset <- method_comparisons(
  data    = ehr500,
  methods = c("Summary_stat", "Linear_mixed_model",
              "Inverse_intensity_rate_ratio_weighting",
              "Joint_modeling_visiting_and_longitudinal_Liang",
              "CIMEHR"),
  method_args = list(
    Summary_stat = list(
      outcome = "log_HbA1c",
      formula = ~ Age + Gender + Marital + Black + Hispanic + NSES
    ),
```

```
    Linear_mixed_model = list(
      fixed = out_form, random = ~ 1 | id, obs_indicator = "R"
    ),
    Inverse_intensity_rate_ratio_weighting = list(
      outcome = "log_HbA1c", visit_covs = covars_all
    ),
    Joint_modeling_visiting_and_longitudinal_Liang = list(
      outcome = "log_HbA1c",
      visit_covs = covars_all, long_covs = covars_all, random_covs = "NSES"
    ),
    CIMEHR = list(
      y_col                        = "log_HbA1c",
      covars_visit_XV              = covars_all,
      covars_outcome_fixed_XY      = covars_all,
      covars_outcome_random_link_ZY = "NSES",
      covars_obs_fixed_XO          = covars_all,
      covars_obs_random_link_ZO    = "NSES"
    )
  ),
  printSE = TRUE,
  print95CI = TRUE,
  true_value_verbose = TRUE
)

print(cmp_subset)
cmp_subset$tables$outcome
```

If a method fails, `method_comparisons()` keeps going by default, stores the error in `cmp$errors`, and prints `NA` rows for that method.

8. Bootstrap inference

`CIMEHR()`, `CIMEHR_timevarying_ou()`, `EHRJoint()`, and `Joint_modeling_visiting_and_longitudinal_Liang()` already return analytic standard errors (sandwich or model-based). `bootstrap()` is recommended for the methods that do not (`CIMEHR_timevarying_integral()`, `Inverse_intensity_rate_ratio_weighting()`, `Inverse_intensity_rate_ratio_balancing()`), and as an optional robustness check or for propagating Stage 1 nuisance uncertainty for the methods that do. The bootstrap resamples subjects with replacement and refits the selected method on each replicate.

```
bs <- bootstrap(
  fit_iirr,
  data = ehr500,
  B    = 200,
  seed = 1
)
print(bs)
summary(bs)
```

For CIMEHR-family methods, forward covariate specifications through `covars`:

```
bs_cimehr <- bootstrap(
  fit_cimehr,
  data = ehr500,
  B    = 200,
  seed = 1,
  covars = list(
    covars_visit_XV               = covars_all,
    covars_outcome_fixed_XY       = covars_all,
    covars_outcome_random_link_ZY = "NSES",
    covars_obs_fixed_XO           = covars_all,
    covars_obs_random_link_ZO     = "NSES"
  )
)
```

9. Simulating custom data

`sim_data_gen()` accepts a `covariates` list spec; per-process coefficient vectors are keyed by the names declared there. The default spec is the generic two-covariate `Z`/`X` form used in the package's internal tests, but the EHR-style spec used to build `sim_ehr_data` is an instructive example:

```
ehr_covs <- list(
  Age      = list(type = "continuous", dist = "uniform",
                  min = 18, max = 99, standardize = TRUE),
  Gender   = list(type = "binary", prob = 0.5),
  Marital  = list(type = "binary", prob = 0.4),
  Black    = list(type = "binary", prob = 0.30),
  Hispanic = list(type = "binary", prob = 0.20),
  NSES     = list(type = "continuous", dist = "normal",
                  mean = 0, sd = 1, standardize = FALSE)
)

dat <- sim_data_gen(
  n          = 500,
  seed       = 42,
  covariates = ehr_covs,
  gamma = c(intercept = -2.0, Age = 0.10, Gender = -0.10, Marital = -0.05,
            Black = 0.15, Hispanic = -0.05, NSES = 0.12),
  alpha = c(intercept = 0.5, Age = 0.10, Gender = -0.10, Marital = 0.00,
            Black = 0.05, Hispanic = -0.05, NSES = 0.20),
  obs_random_covs = "NSES",
  delta   = c(intercept = -0.3, NSES = -0.5),
  sigma_q = c(intercept =  0.3, NSES =  0.2),
  beta = c(intercept = -2.0, Age = 0.05, Gender = 0.10,
           Marital = -0.03, Black = 0.15, Hispanic = 0.05,
           NSES = -0.25),
  outcome_random_covs = "NSES",
  theta   = c(intercept = 0.5, NSES = 0.30),
  verbose = FALSE
)
```

```
head(dat$long_data)
#>   id        Age Gender Marital Black Hispanic       NSES  Age_raw      time R
#> 1  1  1.4509383      0       1     0        1 -1.5515833 92.09929 29.120782 0
#> 2  2  1.5269230      0       0     1        0 -0.3881794 93.90311  6.860926 1
#> 3  3 -0.6941156      1       1     0        0  1.7641481 41.17730 19.518167 0
#> 4  3 -0.6941156      1       1     0        0  1.7641481 41.17730 38.586881 1
#> 5  4  1.1631015      1       0     1        1 -0.4165277 85.26626  2.696261 1
#> 6  4  1.1631015      1       0     1        1 -0.4165277 85.26626  4.649411 1
#>           Y  C
#> 1        NA 60
#> 2 -3.014284 60
#> 3        NA 60
#> 4  5.253490 60
#> 5 -1.823368 60
#> 6 -3.760622 60
```

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